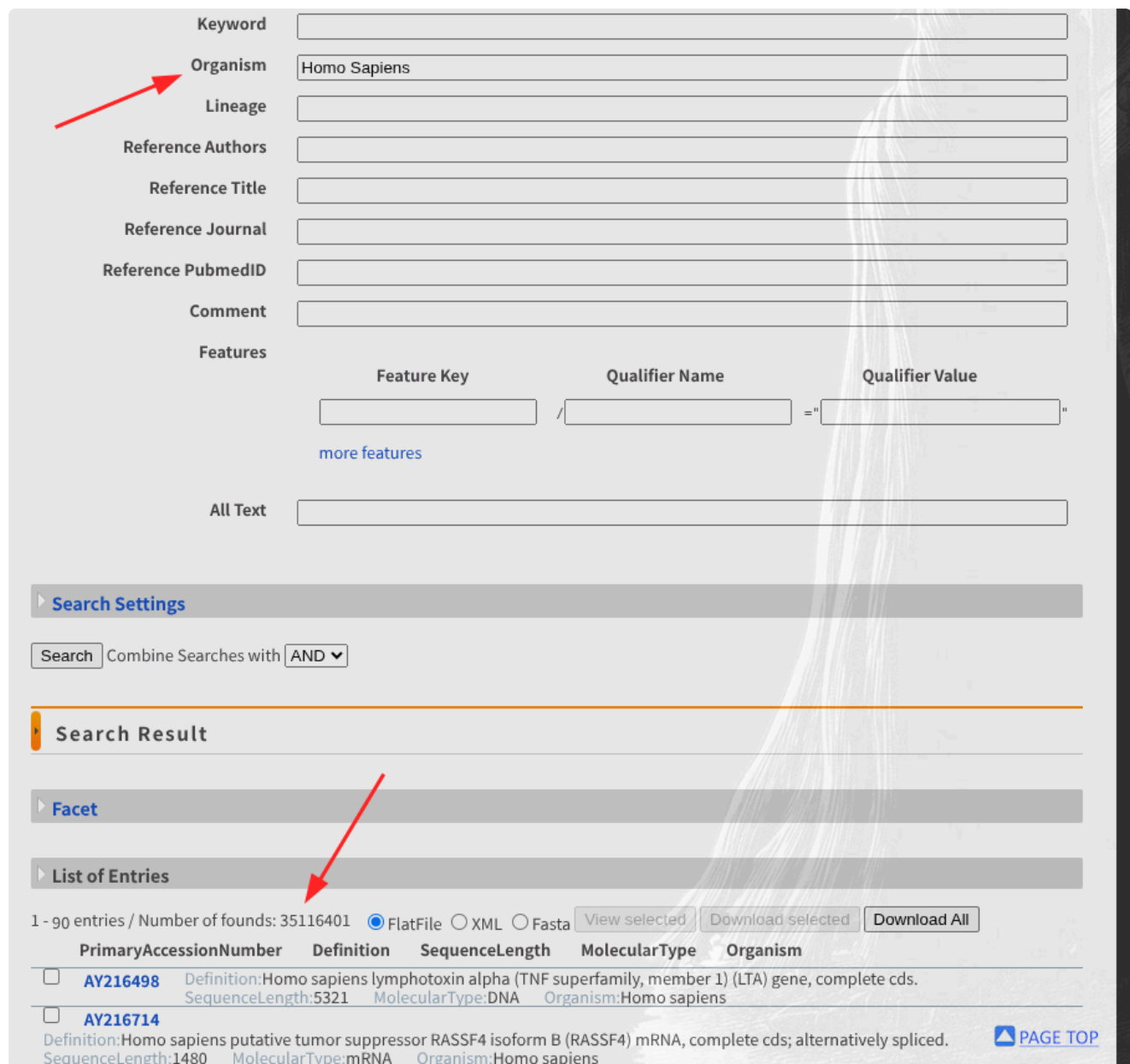


Solution - 2

Q1

DDJB

this link opens the DDJB advanced search for Homo sapiens



Keyword

Organism

Lineage

Reference Authors

Reference Title

Reference Journal

Reference PubmedID

Comment

Features

Feature Key	Qualifier Name	Qualifier Value
	/	= " "

[more features](#)

All Text

Search Settings

Search Combine Searches with

Search Result

Facet

List of Entries

1 - 90 entries / Number of founds: 35116401 ☒ FlatFile ☐ XML ☐ Fasta

PrimaryAccessionNumber	Definition	SequenceLength	MolecularType	Organism
☐ AY216498	Definition:Homo sapiens lymphotoxin alpha (TNF superfamily, member 1) (LTA) gene, complete cds.	SequenceLength:5321	MolecularType:DNA	Organism:Homo sapiens
☐ AY216714	Definition:Homo sapiens putative tumor suppressor RASSF4 isoform B (RASSF4) mRNA, complete cds; alternatively spliced.	SequenceLength:1480	MolecularType:mRNA	Organism:Homo sapiens

[PAGE TOP](#)

so DDBJ = 35,116,401

genbank

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

Nucleotide Homo Sapiens[Organism] [Create alert](#) [Advanced](#) [Help](#)

Species: Animals (29,784,649), Plants (1), Fungi (1), Viruses (4,548), Customize ...

Molecule types: genomic DNA/RNA (16,937,4), mRNA (9,950,426), rRNA (788), Customize ...

Source databases: INSDC (GenBank) (29,256,885), RefSeq (424,468), Customize ...

Sequence Type

Summary 20 per page Sort by Default order

Items: 1 to 20 of 29841998

<< First < Prev Page 1 of 1492100 Next > Last >>

☐ [Homo sapiens N-sulfoglucosamine sulfohydrolase \(SGSH\), RefSeqGene on chromosome 17](#)
26,661 bp linear DNA
Accession: NG_008229.2 GI: 2310189229
[Protein](#) [PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens FASTK mitochondrial RNA regulator 3 \(FASTKD3\), transcript variant 3, non-coding RNA](#)
1,001 bp linear transcribed-RNA
Accession: NR_073608.2 GI: 1890392847
[PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

Filters: [Manage Filters](#)

Send to: **Results by taxon**

Top Organisms [\[Tree\]](#)
Homo sapiens (29781320)
synthetic construct (57325)
Mus musculus (3176)
Human papillomavirus 16 (78)
Trypanosoma cruzi (21)
All other taxa (78)
[More...](#)

Find related data

Database:

Search details

in the nucleotide database in genbank it shows **29,841,998**

EMBL

this link

ebi.ac.uk/ena/browser/advanced-search

Relaunch to update

Data Type Query Inclusion/Exclusion Fields Data Filters Results

Download ENA records: FASTA Download report: JSON

TEXT TSV

Accession	Description
A00383	H.sapiens NANB-hepatitis associated DNA
A06851	H.sapiens gene for relaxin, exon 1
A08002	Nucleotide sequence 7 from patent number EP0370719
A08905	H.sapiens (haplotype 2B, allele MS32, isolate English, serial number 31) minisatellite sequence
A11954	H.sapiens mRNA fragment for calcitonin
A14571	(2'-5') oligo A synthetase cDNA for 1.6kb RNA (E16,clone 9-21)
A25971	H.sapiens mRNA for T-cell receptor IGR b 20 Vbeta 9
A26249	Human shIL5R-alpha gene
A26437	Human granzyme B
A28104	Human GABAA receptor alpha-5 subunit

Items per page: 10 1 - 10 of 10000 (29244316 total)

New Search Back Copy Curl Request Copy Http Request

so **29,244,316**

finally

DDBJ = 35,116,401

GenBank = 29,841,998

EMBL = 29,224,316

Q.2

link

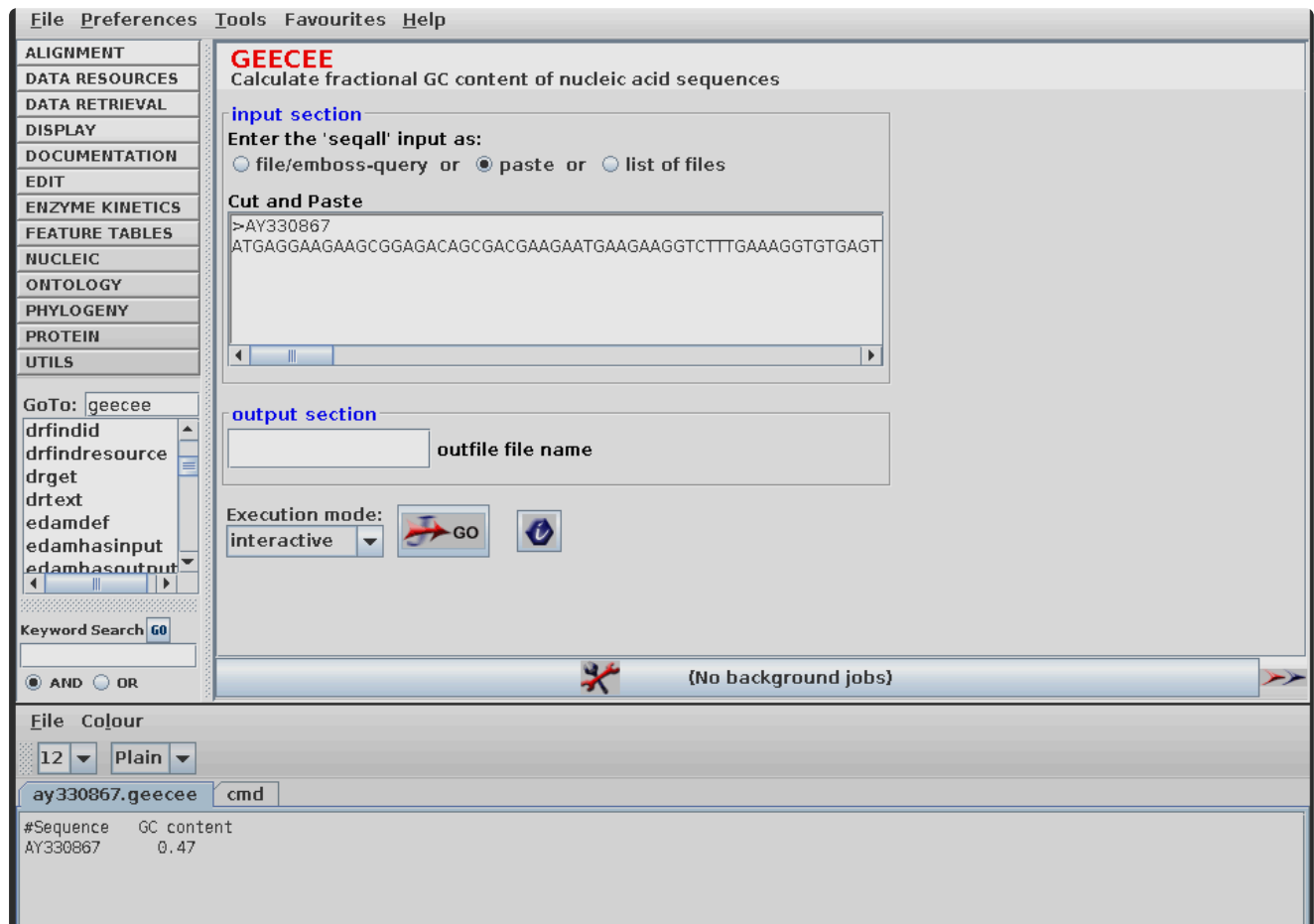
this is the sequence taken from genbank

```

ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTGTGA
GTTGGCCAGAACTC
TGAAAAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACCTGGATGTGT
TTGGCCAAATGGGA
GAGTGGTTACAACACACGAGCTACAACTACAATGCTGGAGACAGAAGCACTGATT
ATGGGATATTTTCAG
ATCAATAGCCGCTACTGGTGTAAATGATGGCAAAACCCCAGGAGCAGTTAATGCCTG
TCATTTATCCTGCA
GTGCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAAGAGGGTTGTC
CGTGATCCACAAGG
CATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGTCCGTCAGT
ATGTTCAAGGTTGT
GGAGTGTAAATGA

```

using emboss we can figure out the GC content



so it is 47%

Q.3

- <https://www.ncbi.nlm.nih.gov/nuccore/33088589>

- https://getentry.ddbj.nig.ac.jp/getentry/na/AY330867/?format=flatfile&filetype=html&trace=true&show_suppressed=false&limit=10
- <https://www.ebi.ac.uk/ena/browser/view/AY330867?show=orcid-links>

all databases give the sequence, translation and other useful info
all three are part of the INSDC and sync daily, so the core biological sequences and accession numbers are identical across all of them
(taken from ddbj)

LOCUS	AY330867	432 bp	mRNA	linear	SYN 29-JUL-2004
DEFINITION	Synthetic construct human lysozyme mRNA, complete cds.				
ACCESSION	AY330867				
VERSION	AY330867.1				
KEYWORDS	.				
SOURCE	synthetic construct				
ORGANISM	synthetic construct				
	other sequences; artificial sequences.				
REFERENCE	1 (bases 1 to 432)				
AUTHORS	Xie,X.D., Chen,Z.H. and Zhou,C.S.				
TITLE	Direct Submission				
JOURNAL	Submitted (26-JUN-2003) Postdoctoral Scientific Research Station of Gansu Yasheng Industrial (Group) Co., Ltd., Zhangye Road, Lanzhou, Gansu 730030, P.R.China				
FEATURES	Location/Qualifiers				
source	1..432				
	/organism="synthetic construct"				
	/mol_type="mRNA"				
	/db_xref="taxon:32630"				
CDS	1..429				
	/codon_start=1				
	/transl_table=11				
	/product="human lysozyme"				
	/protein_id="AAP93336.1"				
	/translation="MRKKRRQRRRMKKVFERCELARTLKRLGMDGYRGISLANWMCLA KWESGYNTRATNYNAGDRSTDYGIQINSRYWCNDGKTPGAVNACHLSCSALLQDNIA DAVACAKRVVRDPQGIRAWVAWRNRCQNRGVRQYVQGGCV"				
BASE COUNT	131 a	72 c	129 g	100 t	
ORIGIN	1 atgaggaaga agcggagaca gcgacgaaga atgaagaagg tctttgaaag gtgtgagttg				
	61 gccagaactc tgaaaagatt gggaatggat ggctacaggg gaatcagcct agcaaactgg				
	121 atgtgttttg ccaaatggga gagtgggttac aacacacgag ctacaaacta caatgctgga				
	181 gacagaagca ctgattatgg gatatttcag atcaatagcc gctactggtg taatgatggc				
	241 aaaacccag gagcagttaa tgcctgtcat ttatcctgca gtgctttgct gcaagataac				
	301 atcgctgat ctgtagcttg tgcaagagg gttgtccgtg atccacaagg cattagagca				
	361 tgggtggcat ggagaaatcg ttgtcaaaac agaggtgtcc gtcagtatgt tcaaggttgt				
	421 ggagtgtaat ga				
//					

but they have some differences like

- genbank and ddbj use full words like locus and origin for headers but embl uses two letter codes like id and sq
- ddbj clearly shows the exact base count right before the sequence but genbank skips it completely

- embl web view gives extra stuff like orcid data claims and cross references to other databases
- genbank and ddbj put the sequence numbers on the left side but embl puts them on the far right
- **genbank** it is tied to NCBI, meaning it natively cross-links to PubMed for literature, BLAST for alignments, and RefSeq for curated genomes
- **embl** it is built into the european framework, directly linking nucleotide sequences to Ensembl (genome browsers) and UniProt (protein databases)
- **ddbj** it serves the asian research node and links out to specific functional databases like KEGG for metabolic pathway analysis

the raw sequence data is exactly the same; the real difference is just the web interface and the downstream analytical tools each database connects us to

Q.4

[papers link](#)

The screenshot shows a Google Scholar search results page. The search query is "discrimination of beta barrel membrane proteins". The page displays several search results, each with a title, authors, year, journal, and a brief abstract. The results are sorted by year, with the most recent at the top. The first result is "A Hidden Markov Model method, capable of predicting and discriminating β -barrel outer membrane proteins" by PG Bagos, TD Liakopoulos, IC Spyropoulos, et al., published in BMC in 2004. The second result is "TMBETADISC-RBF: discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles" by YY Ou, MM Gromiha, SA Chen, M Suwa, published in Computational biology and chemistry in 2008. The third result is "Predicting transmembrane beta-barrels in proteomes" by HR Bigelow, DS Petrey, J Liu, D Przybylski, published in Nucleic acids research in 2004. The fourth result is "PRED-TMBB2: improved topology prediction and detection of β -barrel outer membrane proteins" by KD Tsirigos, A Elofsson, PG Bagos, published in Bioinformatics in 2016. The fifth result is "A sequence-profile-based HMM for predicting and discriminating β -barrel membrane proteins" by PL Martelli, P Fariselli, A Krogh, R Casadio, published in ISMB in 2002. The sixth result is "Functional discrimination of membrane proteins using machine learning techniques" by MM Gromiha, Y Yabuki, published in BMC bioinformatics in 2008. The seventh result is "TMB: Hunt on amino acid composition-based method to screen proteomes for β -barrel membrane proteins" by MM Gromiha, Y Yabuki, published in BMC bioinformatics in 2008.

discrimination of beta barrel membrane proteins

Scholar About 20,500 results (0.41 sec) YEAR

A Hidden Markov Model method, capable of predicting and discriminating β -barrel outer membrane proteins [PDF] springer.com
 PG Bagos, TD Liakopoulos, IC Spyropoulos... - BMC ..., 2004 - Springer
 ... membrane proteins are divided into two distinct structural classes, the α -helical membrane proteins and the β -barrel membrane proteins... proteins of that type are located mostly in the cell ...
 ☆ Save Cite Cited by 234 Related articles All 17 versions

[HTML] TMBETADISC-RBF: discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles [HTML] sciencedirect.com
 YY Ou, MM Gromiha, SA Chen, M Suwa - Computational biology and ..., 2008 - Elsevier
 ... discrimination of β -barrel or outer membrane proteins (OMPs) from other folding types of globular and membrane proteins ... have been proposed for discriminating OMPs based on ...
 ☆ Save Cite Cited by 106 Related articles All 7 versions

Predicting transmembrane beta-barrels in proteomes [PDF] oup.com
 HR Bigelow, DS Petrey, J Liu, D Przybylski... - Nucleic acids ..., 2004 - academic.oup.com
 ... β -barrel membrane proteins directly from sequence. One reason is that only very few high-resolution structures for transmembrane β -barrel (TMB) proteins ... protein discrimination ...
 ☆ Save Cite Cited by 219 Related articles All 16 versions

PRED-TMBB2: improved topology prediction and detection of β -barrel outer membrane proteins [PDF] oup.com
 KD Tsirigos, A Elofsson, PG Bagos - Bioinformatics, 2016 - academic.oup.com
 ... The PRED-TMBB method is based on Hidden Markov Models and is capable of predicting the topology of β -barrel outer membrane proteins and discriminate them from water-...
 ☆ Save Cite Cited by 103 Related articles All 12 versions

[PDF] A sequence-profile-based HMM for predicting and discriminating β -barrel membrane proteins [PDF] academia.edu
 PL Martelli, P Fariselli, A Krogh, R Casadio - ISMB, 2002 - academia.edu
 ... In this paper we use the prototypes of the β -barrel membrane proteins crystallized so far for training and testing a method based on HMM. The model is novel and is trained on ...
 ☆ Save Cite Cited by 249 Related articles All 13 versions

Functional discrimination of membrane proteins using machine learning techniques [PDF] springer.com
 MM Gromiha, Y Yabuki - BMC bioinformatics, 2008 - Springer
 ... of 64% in a data set of 1718 membrane proteins. The application of amino acid occurrence ... have discriminated transporters from other α -helical and β -barrel membrane proteins with ...
 ☆ Save Cite Cited by 93 Related articles All 16 versions

TMB: Hunt on amino acid composition-based method to screen proteomes for β -barrel membrane proteins [PDF] springer.com

some examples:

Liu, Q., Zhu, Y., Wang, B., & Li, Y. (2003). Identification of β -barrel membrane proteins based on amino acid composition properties and predicted secondary structure. *Computational Biology and Chemistry*, 27, 355–361. [https://doi.org/10.1016/s1476-9271\(02\)00085-3](https://doi.org/10.1016/s1476-9271(02)00085-3) Cited by: 52

Ou, Y.-Y., Gromiha, M. M., Chen, S.-A., & Suwa, M. (2008). TMBETADISC-RBF: Discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles. *Computational Biology and*

Chemistry, 32, 227–231.

<https://doi.org/10.1016/j.compbiolchem.2008.03.002> Cited by: 106

Tsirigos, K. D., Elofsson, A., & Bagos, P. G. (2016). PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins. *Bioinformatics*, 32, i665–i671.

<https://doi.org/10.1093/bioinformatics/btw444> Cited by: 103

Q.5

Kihara D [Google Scholar link](#)

one paper published in nature journal example

[paper link](#)

they report the results from the first large-scale community-based critical assessment of protein function annotation (CAFA) experiment.

Q.6

[paper link](#)

"Mitochondrial beta-barrel proteins, an exclusive club?"

there are lots of related articles

Similar articles

[Transport proteins \(carriers\) of mitochondria.](#)

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

[Systematic analysis of the twin cx\(9\)c protein family.](#)

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

[Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.](#)

de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.

Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.

PMID: 19523970

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)

Di Lisa F, Bernardi P.

J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

[See all similar articles](#)

Total 27 similar articles are listed [here](#)

Q.7

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

PubMed

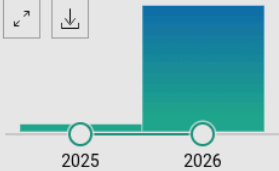
Nature[Journal] AND 2026[Date - Publication] × **Search**

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[Save](#) [Email](#) [Send to](#) Sort by: [Best match](#) [Display options](#)

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[Edit custom filters](#)

RESULTS BY YEAR



PUBLICATION DATE

☒ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

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- ☐ **Ketogenic diet mediates intestinal tumorigenesis through lipids not ketones.**
 1 Shay JES, Chi F, Tzouanas CN, Han S, Zhang X, Ten Hoeve J, Williams KJ, Neptun S, Sever T, Fuentes I, Bhatia SN, Calibasi-Kocal G, Vander Heiden MG, Shalek AK, Yilmaz ÖH.
 Cite **Nature.** 2026 Aug;656(8127):474-483. doi: 10.1038/s41586-026-10779-y. Epub 2026 Jul 15. PMID: 42457955
- ☐ **GPC3-specific dnTGFβRII-armoured CAR T cells for hepatocellular carcinoma.**
 2 Zhang Q, Fu Q, Shen Y, Cao W, Jin G, Zhang Y, Wu J, Chen C, Wang H, Xu X, Sun K, Xue X, Bondanza A, Cao D, Chu N, Durham N, Cobbold M, Farsaci B, Giardino Torchia ML, Hao Y, Hong Y, Huang J, Humphries M, Jian Q, Moody G, Stone J, Sun L, Tu E, Wang F, Wang F, Yang Y, Yao Y, Zou A, Bai X, Liang T.
 Cite **Nature.** 2026 Jul 15. doi: 10.1038/s41586-026-10786-z. Online ahead of print. PMID: 42457964
- ☐ **Diet-microbiome synergy underlies obesity-associated immunotherapy efficacy.**
 3 Desharnais L, Swaby A, Messaoudene M, Doré S, Yu MW, Fiset B, Breton V, Ponce M, Hu Y, Wilson L, Sorin M, Wang Y, Dewar K, Pollak M, Elkrief A, Routy B, Walsh LA, Quail DF.
 Cite **Nature.** 2026 Jul 8. doi: 10.1038/s41586-026-10750-x. Online ahead of print. PMID: 42420462
- ☐ **Mitochondrial metabolism and epigenetic crosstalk drive SASP.**
 4 Martini H, Birch J, Marques EDM, Vittorelli S, Lannado AB, Pirius N, Franco AC, Lee G, Han Y, Rowsay II

Pubmed shows 2937 articles

Search Sources SciVal ? [Create account](#)

Advanced query ☐

EXACTSRCTITLE ("Nature") AND PUBYEAR = 2026

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22,924 documents found [Analyze results](#)

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Search within results

Filters

Year

☒ Range ☐ Individual

from to

☐ All [Export](#) [Download](#) [Citation overview](#) [More](#) [Show all abstracts](#) Sort by [Date \(newest\)](#) [Grid](#) [List](#)

	Document title	Authors	Source	Year	Citations
☐ 1	Article • Open access Minimalist terahertz wireless transceiver in integrated photonics	Guo Y., Zhang X., Zhang Y., ... Wang X., Shu H.	Nature Communications , 17(1), 4429	2026	0
	Show abstract Full Text View at Publisher Related documents				
☐ 2	Article • Open access CXCR3 is associated with T-cell-induced heart damage in acute rheumatic fever	Middleton F.M., McGregor R., Lorenz N., ... Carapetis J., Moreland N.J.	Nature Communications , 17(1), 4664	2026	0
	Show abstract Full Text View at Publisher Related documents				

SCOPUS shows 22,924 articles

Q.8

Burkhard Rost [FOLLOW](#) [GET MY OWN PROFILE](#)

Professor of Computer Science, [Technical University Munich](#)
Verified email at tum.de - [Homepage](#)
[machine learning](#) [artificial intelligence](#) [bioinformatics](#) [computer science](#) [protein prediction](#)

TITLE	CITED BY	YEAR
The transcriptional landscape of the mammalian genome P Carninci, T Kasukawa, S Katayama, J Gough, MC Frith, N Maeda, ... science 309 (5740), 1559-1563	4251	2005
Prediction of protein secondary structure at better than 70% accuracy B Rost, C Sander Journal of molecular biology 232 (2), 584-599	3970	1993
ProtTrans: toward understanding the language of life through self-supervised learning	3809 *	2021

Cited by [VIEW ALL](#)

	All	Since 2021
Citations	65146	19704
h-index	116	60
i10-index	292	182

Citations = 65146

h-index = 116 (highest i have seen for any researcher till now!)

Q.9

link to brenda

Information on EC 1.7.2.3 - trimethylamine-N-oxide reductase

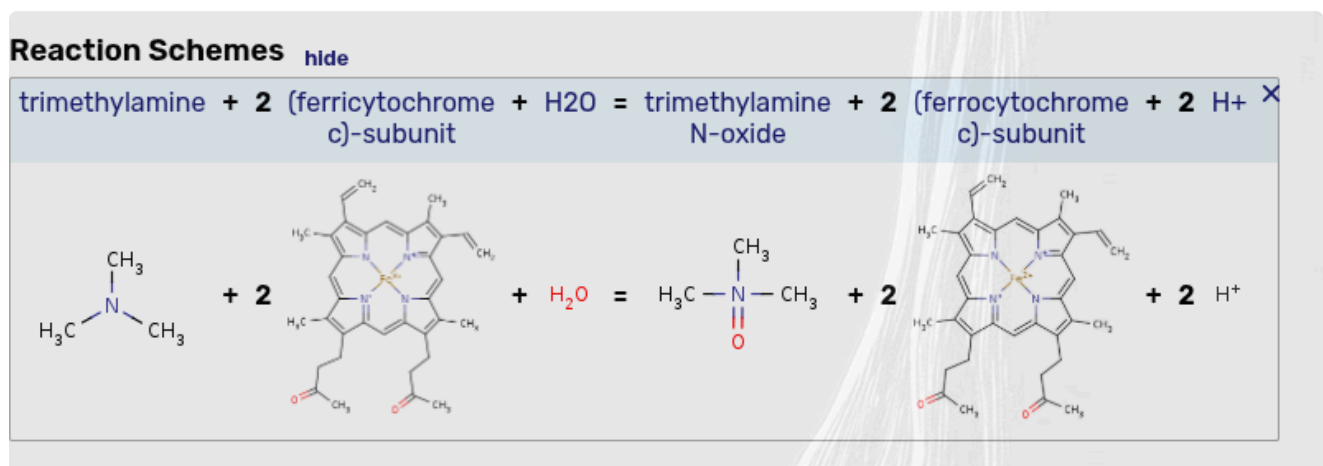
for references in articles please use BRENDA:EC1.7.2.3

EC Tree → 1 Oxidoreductases → 1.7 Acting on other nitrogenous compounds as donors → 1.7.2 With a cytochrome as acceptor → 1.7.2.3 trimethylamine-N-oxide reductase

this shows the enzyme class hierarchy

- **Accepted Name:** Trimethylamine-N-oxide reductase (cytochrome c), also commonly known as TMAO reductase.
- **Enzyme Class Hierarchy:**
 - **EC 1 (Class):** Oxidoreductases (enzymes that catalyze oxidation-reduction reactions).
 - **EC 1.7 (Subclass):** Acting on other nitrogenous compounds as donors.
 - **EC 1.7.2 (Sub-subclass):** With a cytochrome as an acceptor.
- **Function / Catalytic Activity:** This microbial enzyme catalyzes the chemical reduction of trimethylamine N-oxide (TMAO) into trimethylamine (TMA) as part of the electron transport chain.

reaction



Q.10

finding the EC number on brenda

 **6.3.5.4** asparagine synthase (glutamine-hydrolysing)

now searching it on ebi(thornton) gives us the enzyme

<https://www.ebi.ac.uk/thornton-srv/m-csa/entry/302/>

[1ct9Q](#) - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å)  

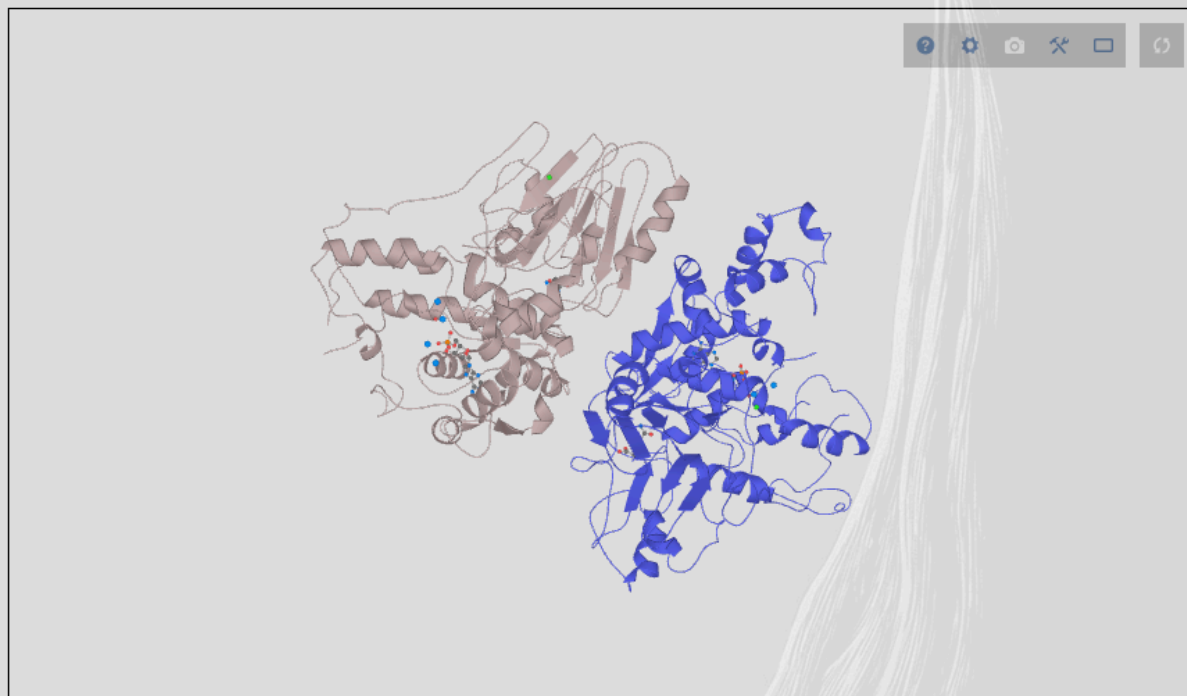


Catalytic CATH Domains

[3.60.20.10Q](#)  [3.40.50.620Q](#)  (see all for [1ct9Q](#))

Cofactors

Magnesium(2+) (1) [Metal MACiE](#)



Show Active Center

and below it shows the catalytic residues and their roles

Catalytic Residues Roles			
UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

so answer = Cys 1, Cys 2, Gly76, Leu 51, Thr 322, and Arg 325.

Q.11

Search for as complete name

Homo sapiens

Taxonomy ID: 9606 (for references in articles please use ncbitaxon:9606)

current name

Homo sapiens Linnaeus, 1758

Genbank common name: **human**

NCBI BLAST name: **primates**

Rank: **species**

Genetic code: [Translation table 1 \(Standard\)](#)

Mitochondrial genetic code: [Translation table 2 \(Vertebrate Mitochondrial\)](#)

Lineage(full)

[cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Superstomi](#)

just listing the data below now

Organism	Scientific Name	Taxonomy ID	Number of Chromosomes (2n)
Human	<i>Homo sapiens</i>	9606	46
Cat	<i>Felis catus</i>	9685	38
Dog	<i>Canis lupus familiaris</i>	9615	78
Domestic guinea pig	<i>Cavia porcellus</i>	10141	64
Thale cress	<i>Arabidopsis thaliana</i>	3702	10

Q.12

The **Entrez Programming Utilities (E-utilities)** are a suite of server-side programs that serve as the public API for the NCBI Entrez system. They allow users to programmatically search, link, and retrieve data from across all NCBI Entrez databases (including PubMed, Nucleotide, Protein, and Gene) without needing to use the web browser interface.

example syntax

```
esearch.fcgi?db=<database>&term=<query>
```

example

- ```
[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science[journal]+AND+breast+cancer+AND+2008[pdat]]
(https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science%5bjournal%5d+AND+breast+cancer+AND+2008%5bpdat%5d)
```

to fetch a fasta format

```
[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi
?
db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text
]
```

syntax is

- **db=nuccore**: Specifies the database (e.g., **nuccore** for nucleotide, or **protein**).
- **id=AY330867**: The unique identifier or accession number of the record.
- **rettype=fasta**: Tells the server you want the specific FASTA layout.
- **retmode=text**: Tells the server to return plain text rather than XML.

## Q.13

### a. Protein properties

1. **UniProtKB (Universal Protein Resource)**: The central hub for the collection of functional information on proteins, with accurate, consistent, and rich annotation of protein properties.
2. **Pfam (or InterPro)**: A large collection of protein families, represented by multiple sequence alignments and hidden Markov models (HMMs), used to determine protein domain properties.  
*\*b. Small molecules (Structure related)*
3. **PubChem**: A massive database maintained by NCBI that provides information on the chemical structures, physical properties, and biological activities of small molecules
4. **ChEBI (Chemical Entities of Biological Interest)**: A freely available dictionary of molecular entities focused on strictly "small" chemical compounds and their structural properties.

### c. Cancer gene databases



5. **COSMIC (Catalogue of Somatic Mutations in Cancer):** The world's largest and most comprehensive resource for exploring the impact of somatic mutations in human cancer.
6. **cBioPortal for Cancer Genomics:** An open-access resource for interactive exploration of multidimensional cancer genomics data sets.